

Turbulence modulation via dilute surfactant additives in experimental Taylor–Green flow

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Abstract here.

I. INTRODUCTION

Since the discovery that trace amounts of dissolved polymer could lead to drag reduction in pipe flows [1], the use of additives for drag reduction in wall bounded flows has become established. Besides polymer solutions, it has since been discovered that self-assembling surfactant solutions share similar effects [2], owing to the fact that dilute surfactant solutions can form wormlike micelles whose relaxation dynamics mimic that of polymer chains while resisting mechanical degradation. Beyond the applications of drag reduction, it has shown that viscoelastic additives are known to modulate turbulence more broadly, reshaping the energy spectrum and producing an elastic range governed by additive elasticity rather than inertial at the small scales [3, 4], with surfactant-specific micellar structures shown to alter vortex shedding and turbulence statistics directly [5].

In spite of academic and commercial interest, comparatively few studies address the behavior of dilute surfactant solutions in the context of three-dimensional bulk turbulence, with most of the literature focused on wall-bounded or quasi-two-dimensional flows. This is in part related to the fact that the rheology of very dilute surfactant solutions remains difficult to pin down, with reported values of viscosity and relaxation time of CTAC/NaSal solutions (such as the considered in the present experiment) varying considerably between studies [6, 7]. In spite of the diverging measurements and reports that some of the dilute solutions behave as a Newtonian fluid under rheological measurement [6], such solutions are known to produce measurable drag reduction [8], which further reinforces the need to analyze such solutions more in depth.

Motivated by this gap, here we present an experimental analysis of the turbulence generated by a four-rotor, Taylor–Green-like experiment using dilute CTAC solutions, in an attempt to characterize the modulation of turbulence caused by the addition of the surfactant across a wide range of Reynolds numbers for flows driven by inertial forcing.

II. EXPERIMENTAL DEVICE

In the present study, we employ an experimental device designed to produce Taylor–Green-like forced turbulence on a laboratory setting. The apparatus, with schematic shown on figure 1, consists of a water tank

with four rotors which are driven by a gear system and rotate in opposing directions in order to produce turbulence in the region between them (hereinafter simply denominated bulk), which is the target of our studies. Spatially-resolved measurements are obtained via 2D particle tracking (briefly described in section II A) via a traditional laser-sheet and high-speed camera combination, which are positioned as also shown in figure 1 for the case of bulk imaging.

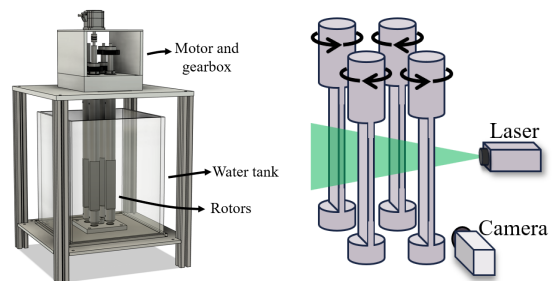


FIG. 1. CAD of the experimental device (left) and schematic showing normal orientation of camera and laser sheet, as well as direction of rotation of the rotors.

Aside from the imaging of the bulk region, it is also possible to change the relative orientations of the laser sheet and camera, and use a mirror to allow for top-down imaging of the region in the vicinity of the blades, as shown in the schematic of figure 2.

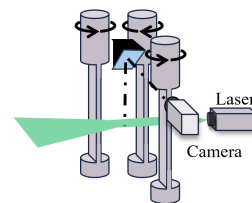


FIG. 2. Schematic of the altered orientation of camera and laser using mirror for imaging of the space between rotors. One of the rotors is omitted for clarity.

As shown in the schematics of figures 1 and 2, the rotors themselves are shaped as blades with rectangular cross-section. While the design of the experimental device allows for the geometry of the rotors to be changed between cylinders and the carved blades, the use of the latter allows us to isolate inertial forcing in the driving of the flow, and therefore ignore boundary-layer effects which would be present in the case of viscous driving

which happens when cylindrical rotors are used. Additionally, the rotation direction is such that mean stretching of the flow happens in the out-of-plane component of the camera image (when imaging the bulk region in the vertical plane), and is accompanied by the formation of secondary vortices as shown in previous numerical studies [9]. For simplicity, the carved rotors will hereinafter simply be denominated blades.

While not shown in the schematics, the experimental device is also encased in thermal insulation and connected to an air-processor system. By controlling the temperature of the surrounding air, the water temperature was maintained at approximately 26 °C over the course of the experiments.

A. Data measurement and reduction

As previously mentioned, processing of the obtained images to calculate the velocity fields is conducted via a particle tracking velocimetry (PTV) code developed in-house. Although the details of the algorithm are out of the scope of the current document, the code is built as a collection of previously established methods. For the detection of tracer particles from the images, we employ a variation of the Iterative Particle Reconstruction (IPR) method, which is commonly employed for volumetric particle tracking as part of the Shake-the-Box method [10]. The tracking of detected particle positions between the video frames is carried out by a variation of the Polar Coordinate-System Similarity (PCSS) method [11].

Before the calculation of the final velocity vectors through finite differencing, the trajectories are filtered according to a Wiener Filter [12] which is calculated from the spectra of position of the obtained trajectories. The filtering is done in order to avoid the effect of the noise floor in the calculation of the velocities. (need to revisit this) (also need to mention the interpolation that is done before the spatial average) **also supplement this with the explanation of the uncertainty calculation and all.**

B. Surfactant solutions

The surfactant used in the current study was the cationic surfactant cetrimonium chloride (CTAC), with the addition of sodium salicylate on a molar ratio of approximately 4:1 to provide counterions. The solution is prepared by mixing the adequate amounts of surfactant and salt with tap water and then agitating in a magnetic stirrer over a period of at least 12 hours, after which the resulting solution is added to the experimental device. For simplicity, the surfactant and salt solution are hereinafter denominated simply as the CTAC solution, with the solution concentration corresponding exclusively to the concentration of the surfactant counterpart.

For the analysis of the effect of solution concentration on the results, three CTAC solutions of concentrations of

33 ppm, 50 ppm, and 100 ppm were prepared. As previously mentioned, rheological measurements of such dilute solutions have historically yielded diverging results [6, 7], meaning that to the best of the author's knowledge, there is no precise measurement of the properties such as the relaxation time of micellar structures under these conditions.

Moreover, the molar ratio of 4:1 chosen for the current study differs from that of the aforementioned measurements, as well as that of drag reduction studies [8]. Although numerical studies have reported that such high molar ratios can lead to the formation of branched structures and loss of viscoelasticity [13], such claims have been made for solutions of much higher concentrations. In the experience of the authors, this effect is not as pronounced in the case of dilute solutions, and the use of a high molar ratio has been chosen to favor the formation of longer micellar structures.

In spite of these divergences, it can be expected from the aforementioned studies that this relaxation time should be in the order of 10^{-1} seconds. Since the relaxation time cannot be precisely known beforehand or independently controlled over a wide range, a sweep over the Weissenberg **you need to define what is the weissenberg number** **bro** number is instead achieved by changing the Reynolds number of the generated turbulence. For each of the prepared solutions, the rotation rate of the blades is changed 2^{-3} to 2^4 Hz in powers of two. Exclusively for the 100 ppm solution **say how the rotation rate was reduced further**. Although it is debatable whether the flow generated in the case of the lower rotation rates can be denominated turbulent, it remains chaotic and unsteady, so such cases are still useful in the determination of the effect of surfactant addition on such unsteady but low Reynolds number types of flows.

Note that although it can be expected that the precise value of τ_s should change with the concentration of the solutions, the results of the aforementioned studies indicate that this change is not necessarily monotonic. Nevertheless, the same results indicate that the variations in relaxation time should be small enough for the precise value to remain in the same order of magnitude. For this reason, we take the representative value of $\tau_s = 0.1$ s as the boundary at which $Wi \approx 1$ for the three solutions. With this consideration, the range of Weissenberg numbers are obtainable in the current experiment range from $Wi = \mathcal{O}(10^{-2})$ to $Wi = \mathcal{O}(10^2)$, allowing us to see the transitions to the attenuated regime regardless of small changes in the precise value of relaxation time.

The next section describes briefly the characterization of the turbulence generated in the current experimental device for the case of Newtonian fluid.

III. TURBULENCE CHARACTERIZATION

Considering the unique nature of the current experimental apparatus, we conduct a characterization of the

turbulence generated in the case of Newtonian flow as a function of the rotation rate of the blades. Although the generated turbulence is anisotropic, turbulent statistics become approximately homogeneous in the vicinity of the stagnation point of the mean flow in the center of the experiment. Owing to this, the statistics are computed as spatiotemporal averages over the duration of the experiments. Figure 3 displays the velocity fluctuation RMS as a function of the blade rotation speed, demonstrating that there is an approximately linear trend, with the ratio between the two components remaining somewhat constant as well. Note that the turbulent intensity is more pronounced in the horizontal direction, consistent with the primary forcing direction of the blades.

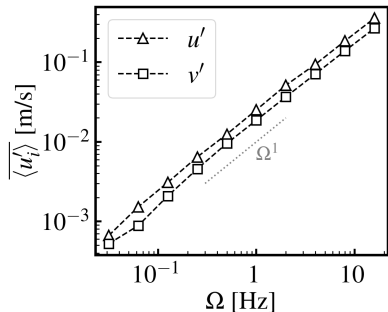


FIG. 3. Spatially averaged velocity fluctuation RMS in the bulk as a function of rotation rate of the agitator blades.

Although direct measurement of the dissipation range is not feasible under current experimental conditions, the turbulent dissipation can be estimated through the third-order structure function, as shown in figure 4. Note that although the generated turbulence is anisotropic, the results of figure 4 indicate that the dissipation rate is approximately independent of the measurement direction, which helps to validate the obtained results **is this correct?**. In all the plots, the increase in the value of the structure function as the separation distance reaches the smallest values is caused by the finite depth-of-correlation and the nature of planar imaging, as well as the fact that no interpolation of the instantaneous velocity fields is used for the calculation of these statistics.

For the cases of high rotation rate, the appearance of a horizontal region indicates that an inertial range is well-defined, and the dissipation rate can be taken as the average across this region. For the lower rotation rates, the Reynolds number is too low for the appearance of a defined inertial range, and the extraction of a representative value of dissipation is more difficult. For consistency between the cases, we consider the average value of the normalized third-order structure function between the separations of $r = 3$ mm and $r = 20$ mm as the dissipation value. These representative values are represented as the dotted lines of figure 4.

From these results, we can then obtain the approximate Kolmogorov lengthscale and timescale for each of the cases, as shown in figure 5. From these results, the es-

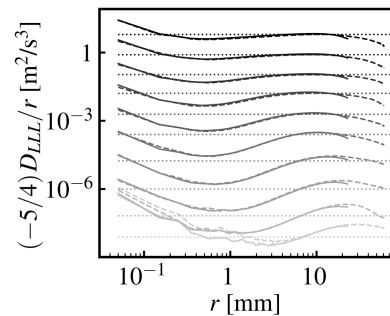


FIG. 4. Normalized third-order structure function for different rotation rates. Darker lines represent a higher rotation rate.

timated Kolmogorov lengthscale provides us with an additional clue as to why it is not possible to directly measure the dissipation range, considering that these scales become smaller than the depth-of-correlation of the current imaging system for most rotation rates. Additionally, wide range of the estimated timescales indicate that a wide range of Weissenberg numbers should be obtained regardless of the fact that the relaxation time of the surfactant solutions is not precisely known.

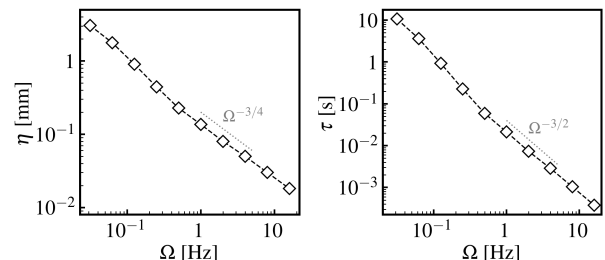


FIG. 5. Kolmogorov lengthscale and timescale as a function of rotation rate of the agitator blades.

IV. RESULTS

The analysis of turbulence modulation by surfactant addition begins with the turbulent intensity at the bulk region, which is described here by the spatially averaged RMS of the velocity fluctuations. Since we conduct planar measurements, this magnitude is calculated from the two measured components, by defining $q'_{2D} = \sqrt{\langle u'^2 \rangle + \langle v'^2 \rangle} = \sqrt{u'^2_{rms} + v'^2_{rms}}$, which is hereinafter referred to as q' for simplicity. The results are then spatially averaged across the measurement region, since these statistics are approximately homogeneous in this area as previously discussed. Note that although the value of q' does not contain the out-of-plane component of the turbulent kinetic energy, its magnitude can be expected to be similar to the currently measured horizontal component of the velocity fluctuations. With this

in mind, the results of q' are interpreted as an indication of the trends of turbulent kinetic energy in the bulk region. Aiming to observe the modulation of such a quantity, figure 6 then presents the ratio of the spatially averaged velocity fluctuation RMS between the surfactant and Newtonian cases.

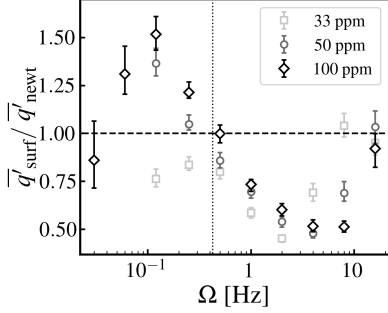


FIG. 6. Ratio of spatially averaged velocity fluctuation RMS between the surfactant and Newtonian cases. Vertical dashed line represents the rotation rate for which $Wi \approx 1$ considering a relaxation time of $\tau_s = 0.1$ s.

According to the classical picture of turbulence modulation via the addition of surfactants and polymers, eddies with a turnover time shorter than the relaxation time of the viscoelastic solutions should be attenuated, and the turbulence in the regime where $Wi < 1$ should remain approximately unchanged. The results shown in figure 6 however, display a more complex picture. It can be seen that although turbulent energy is indeed attenuated once the Weissenberg number crosses the frontier of unity around all cases, the turbulence also appears to be intensified for the 50 ppm and 100 ppm solutions at low rotation rates. Furthermore, once the rotation rate is increased past a certain threshold, the attenuation is lost and the solutions appear to return to the Newtonian behavior.

To help us understand the reason behind these phenomena, We can start by looking at the effect of surfactant addition on mean flow intensity, which can be defined as the average flow velocity at a given region around the stagnation point. Considering the symmetric nature of the experimental apparatus, as well as the forcing with respect to the right and left regions of the measurement domain, we can enforce this symmetry on the mean flow to improve convergence of the results. This is done by defining a mirror coordinate system as $\tilde{\mathbf{x}} \equiv (2x_c - x, y)$, with $x = x_c$ corresponding to the symmetry axis, or stagnation point. This then allows us to write a symmetry-enforced mean flow field component, defined as:

$$U^\dagger(\mathbf{x}) = \frac{1}{2} [U(\mathbf{x}) - U(\tilde{\mathbf{x}})]. \quad (1)$$

We can then compute the spatial average of this quantity to then obtain a representative intensity of the mean flow. Since the field is antisymmetric however, this spatial average is computed only on the left-hand side of

the point (such that $x < x_c$), resulting in the quantity $\bar{U}_L^\dagger$. Note that this approach is chosen (in opposition to taking the spatial average of the velocity magnitude) because it allows us to preserve the flow direction, and thus take into account flow reversal as well should it happen. Moreover, since the defined quantity has the significance of “spatially averaged mean flow” in the bulk region, it will hereinafter simply be denominated as $\bar{U}$ for visual clarity. With this in mind, figure 7 presents the ratio of mean flow intensity between the surfactant and Newtonian cases.

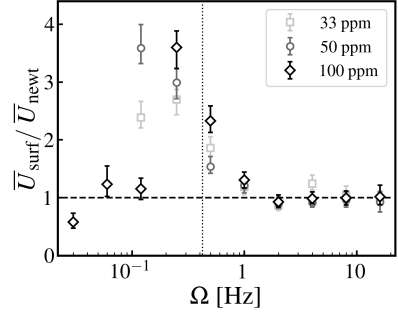


FIG. 7. Ratio of spatially averaged horizontal velocity magnitude between the surfactant and Newtonian cases. Vertical dashed line represents $Wi \approx 1$.

From the plot of figure 7, it can be seen that for rotation rates of approximately 1 Hz or above, the addition of surfactant has almost no effect on the mean flow, irrespective of the concentration. It can be stated that such would be the expected behavior for turbulence which is inertially driven, since the role of surfactant additives is traditionally considered as to attenuate the small scales of the flow. Since the driving rotors act on the large scales, which even at the slowest rotation rate are larger than the Kolmogorov lengthscale by a factor of at least one decade (ref. fig. 5), it can be expected that such large scales should remain mostly unaffected. Nevertheless, the results from figure 7 indicate that reducing the rotation rate leads to an increase of up to 4 times on the mean flow intensity for the regime where $\Omega \lesssim 1$ Hz. Additionally, this behavior appears to be independent of the frontier of $Wi \approx 1$, with the peak in mean flow intensification occurring well below the value of unity of the Weissenberg number, after which the amplification effect appear to cease once the rotation rate is further decreased.

To investigate the cause behind this increase in the mean flow for the lower rotation rates, figure 8 presents the value of the same spatially averaged horizontal component of the mean velocity, but as a function of the phase of the blade (ϕ). Similarly to the previously defined quantities, we introduce $U_\phi^\dagger(\mathbf{x}, \phi) = \frac{1}{2} [U_\phi(\mathbf{x}, \phi) - U_\phi(\tilde{\mathbf{x}}, \phi)]$ corresponding to the symmetry-enforced version of the phase-locked component of the mean flow U_ϕ . Again, for visual clarity, this quantity is

hereinafter simply denominated as $\overline{U}_\phi$ after the spatial averaging over the left side of the observed domain is taken.

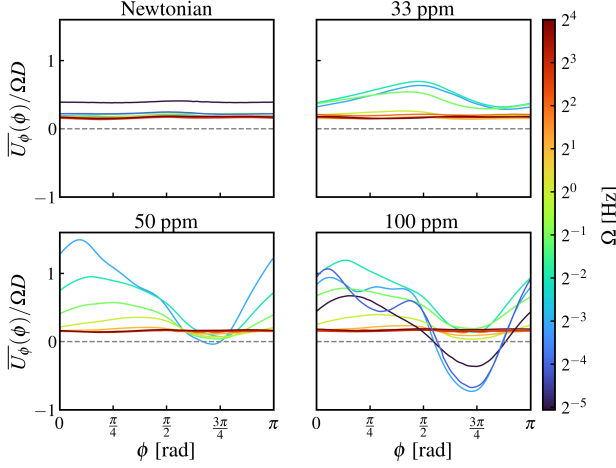


FIG. 8. Spatially averaged horizontal mean flow component as a function of blade phase for each of the tested cases. Positive value refers to inward motion (towards the stagnation point), while a negative value indicates flow reversal.

From these plots, it is possible to see that the overall dependence of the mean flow with the phase of the blade gets amplified as either the concentration of the solution is increased or the rotation rate is decreased, possibly leading to a reversal of flow direction as well. Looking in more detail, from the plots of the 50 and 100 ppm solutions, it can be seen that as the rotation rate is decreased from a high value, the oscillatory behavior is asymmetric, in the sense that we see a strong increase in the mean flow intensity for most of the cycle, and only a small decrease during a very narrow range. This behavior reaches an extreme around the rotation rate of 2^{-2} or 2^{-3} Hz (depending on solution concentration) where the intensification of the mean flow reaches a peak. To aid the analysis, figure 9 presents the normalized amplitude of the curves for each of the cases.

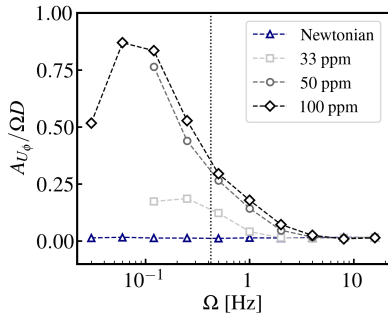


FIG. 9. Normalized amplitude of the spatially averaged, phase-locked horizontal component of the mean flow as a function of rotation rate.

The amplitude curve for each of the solutions indi-

cates that while the Newtonian flows displays an approximately constant normalized amplitude, the surfactant solutions show the presence of a peak, or frequency at which the oscillations are most pronounced. While data for the slowest rotation rates of the 50 ppm solution is not available, analysis of the curve shapes from figure 8 indicates that the oscillations grow in similar manner between the cases, and that the oscillation peak of the 50 ppm solution should happen at a lower frequency than the 100 ppm counterpart.

Returning to the plots of figure 8 once again, it can be seen that for the 100 ppm solution, where the two slowest rotation rates were tested, it is possible to see that as the rotation rate is further reduced, it is possible to see that the oscillations then tend to become more symmetric, with the appearance of a pronounced region of flow reversal. Even though the rotation of the blades is always in the sense of pushing flow towards the central region (as seen from the point of view of the camera), an opposite movement is observed, with the flow going from the center towards the lateral edges of the domain. On the opposite end of the spectrum, it can be seen from both plots that past a certain rotation rate (approximately 4 Hz in the current experiment), the oscillations cease, and the three solutions recover the Newtonian behavior, similarly to what is seen in the plot of figure 7.

The behavior observed in figures 8 and 9 is indicative of a resonance phenomenon between the flow forcing and large structures which have a relaxation time much slower than the typical relaxation time of the micelles, which as previously described, is in the order of 0.1 s. Considering that the current resonant peaks appear at a rotation rate of approximately 0.1 Hz (and therefore forcing frequency of 0.2 Hz due to geometry of the blades), it can be expected that the structures at play in this case also have a relaxation time of the order of 5 s. Moreover, for them to interact with the blades and produce the observed scale of velocity oscillation, the spatial scale of these structures can also be expected to be in the order of $\mathcal{O}(1\text{cm})$, the same as the spacing between the blades.

In order to analyze the movement and behavior of such flow structures in more detail, we can rely on the flow fields measured in the horizontal plane and in the vicinity of the blades. Considering the limited viewing area enforced by the proximity of the blades, and maximum mirror size that can be fitted between them, the schematic shown in figure 2 is slightly modified such that the line of sight of the camera points to the region between two blades, in the orientation at which the motion of “pushing” fluid towards the stagnation point is targeted. In preparation for the analysis of the results, figure 10 shows the relative position of the measured area, displayed as the black square with the velocity vectors. The cross-sections of the blades are drawn as gray rectangles with the rotation direction specified, and the relative positions of the laser sheet and camera viewing direction for the cases of bulk imaging (used for the extraction of the results discussed up to this point) are also shown for refer-

ence. In each of the blades, the points defined as leading and trailing edges are also specified as they will be referred to in the analysis of the results. Note that for clarity reasons only a fraction (15% to 20%) of the tracked particles are shown, and the figures are not representative of the actual spatial resolution of the measurements.

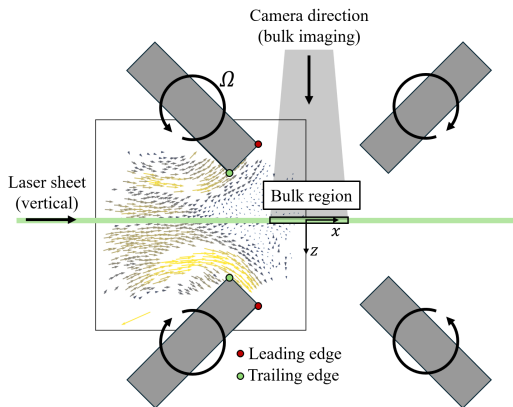


FIG. 10. Schematic of the viewing area in the horizontal plane with respect to the position of the blades, as well as laser sheet and camera directions during bulk imaging.

With this schematic in mind, figure 11 displays the evolution of the flow field over the course of half a rotation period divided in four snapshots, for the Newtonian and 50 ppm cases at the rotation rate of 0.125 Hz divided into the top and bottom rows respectively. In both cases, the evolution is displayed from left to right, with the blades initially in the position of largest constriction of the flow. From there, each consecutive snapshot is delayed by a rotation of 45° . For the Newtonian flow case, in the top row, no particularly noteworthy flow structures are visible, with the highest flow velocity seen in the vicinity of the blades (particularly behind the trailing edge), and a general attenuation of the velocity magnitude towards the bulk region. For the surfactant case in the bottom row however, the more complex flow patterns are observed.

Analyzing the flow structures of the surfactant case, in the bottom row of figure 11, we start at the first snapshot where the blade edges are in the closest approach. In this orientation the motion of the blades is accompanied by broader (when compared to the Newtonian flow) higher velocity regions which extend from the leading edge to the trailing edge direction, and behind the movement of the blades. In the second snapshot, as the two blades begin to separate, these two faster moving regions transform into a more developed shear-layer structure, with an increase in the flow velocity of the layers in direct proximity to the blade and initial formation of secondary high velocity structures in the space between the blades.

At the third snapshot, the layered structures continue to develop, and the velocity magnitude in the region between the blades also reaches the maximum value. Note that the velocity of these streaks, or jet-like structures

surpasses the tip velocity of the blades by over 4 times, and are strongest in the position of largest separation of the blades. It can be hypothesized that as the leading and trailing edges of the blades drag large micellar networks, they are initially compressed as the blades reach closes proximity, and then relax leading to a pronounced acceleration of the flow in this condition. As these “jets” then impinge into the bulk, this region itself then receives lateral compression which relaxes once the jets disappear, leading to the reduction in mean flow intensity and possibly flow reversal in this region.

In the fourth snapshot of figure 11 we can see that the jet-like structures appear to break down or more simply lose strenght in measured region, while an increase of velocity in the top and bottom right portions of the domain can be seen, likely signalling that these structures are now being pulled apart and downwards or upwards for each of the blades. As we return to the first snapshot the constriction of the flow by the two blades rapidly leads to the attenuation of the high velocity regions between the blades, which is also consistent with the rapid drop in velocity which is seen in figure 8.

The streak-like structures shown in figure 11 show resemblance to the elongated structures observed in drag-reduction studies in channel flow using solutions of similar concentrations [14]. In the aforementioned study, the authors argue that the formation of shear-induced-structures (SIS) are a key part of the drag-reducing capability of surfactant, and the spatial measurements of flow velocity in the wall-parallel planes of the drag reducing solutions show similar elongated structures in the vicinity of the walls. In our current experiment, there exists a region of mean shear between the blades due to the velocity gradient up to the bulk region where mean flow is stagnant. It can be hypothesized that this shear, coupled with the relatively low turbulent intensity of the slower rotation rate cases leads to favorable conditions for the formation of SIS, which we see as the jet-like structures of figure 11.

For the slower rotation rates (below 1 Hz), the interaction with blade geometry and forcing frequency leads to a resonant behavior which leads to the acceleration of these structures and an increase in the mean flow. As the shear rate increases further however, their formation is no longer favored (or the large structures break down) and the mean flow intensity returns that of the Newtonian fluid. Moreover, the fact that reducing the rotation rate even further reduces the amplitude of velocity fluctuations is indicative of a mismatch in temporal scales between the rotation rate of the blades and the elasticity of the large structures, which decreases the amplitude of the oscillations and is consistent with the idea of resonance between the large structures and the forcing frequency.

Having established the importance of blade phase on the state of the mean flow in the current experiment, we can also change our analysis of the evolution of the velocity fluctuations. For this, we apply the triple decom-

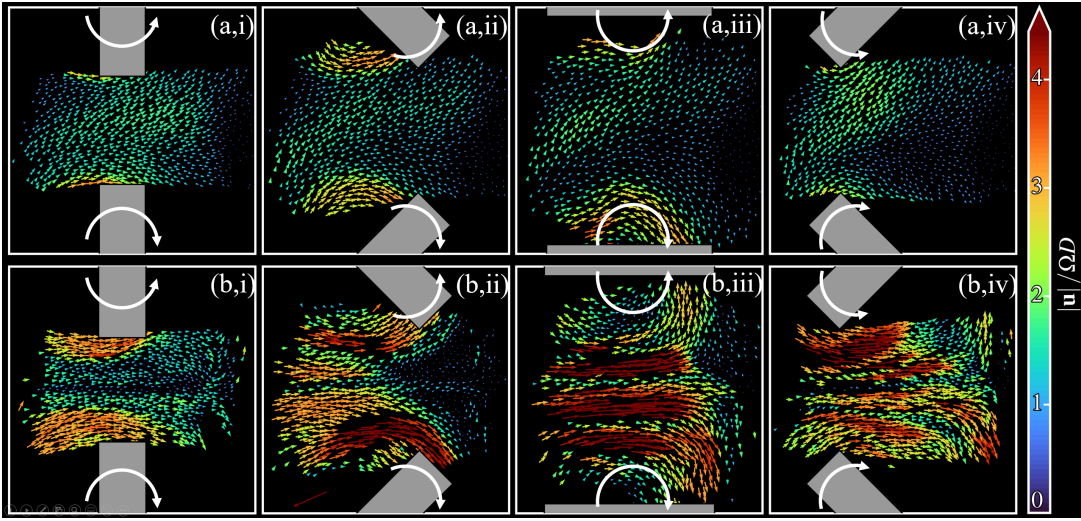


FIG. 11. Top-down visualization of the flow field from Newtonian (a) and 50 ppm (b) cases at rotation rate of 0.125 Hz. For the top and bottom rows, plots (i) through (iv) represent four consecutive snapshots separated by phase difference of $\pi/4$.

position technique, which essentially consists in decomposing the flow field in its mean, periodic, and chaotic components [15], in order to remove the contribution of the periodic forcing which would otherwise be included as part of the velocity fluctuations. Essentially, subtract the phase-dependent mean flow shown, as shown in figure 8 from the instantaneous flow field in order to obtain the chaotic component of the velocity fluctuations, with magnitude simply defined as (2) to keep consistency with previously defined quantities. The ratio of this spatially averaged, phase-corrected RMS between the surfactant and Newtonian cases is then presented in figure 12.

$$q'_\phi = \sqrt{\left\langle (u(\mathbf{x}, t) - U_\phi(\mathbf{x}, t))^2 \right\rangle + \left\langle (v(\mathbf{x}, t) - V_\phi(\mathbf{x}, t))^2 \right\rangle} \quad (2)$$

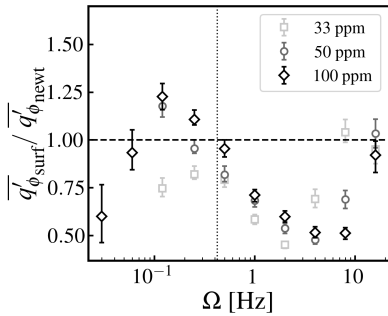


FIG. 12. Ratio of spatially averaged velocity fluctuation RMS (phase-locked) between the surfactant and Newtonian cases.

When contrasted with the results obtained without the triple decomposition (cf. fig. 6), the results of figure 12 show a less pronounced increase in the turbulent energy for low rotation rates, and the pictures becomes closer to what is traditionally expected of turbulence modulation

by surfactant addition. Most notably, we see attenuation of the turbulent energy once the Weissenberg number crossed the frontier of unity (approximately), with an attenuation which is proportional to the rotation rate in the vicinity of this frontier. In this region, the turnover time of the smallest eddies begins to surpass the relaxation time of the string-like micellar structures, which are extended and begin to therefore extract energy from these small scales in the process. As the Reynolds number is increased, more turbulent scales begin to interact with the micelles and the attenuation becomes more pronounced. It can also be said that since the attenuation of the turbulent energy does indeed begin to happen around the point where the Weissenberg number was previously estimated to be around unity, this is a good indicative that the relaxation time of the solutions is indeed in the order of 0.1 seconds. The presence of other effects, however, such as the resonance with large-scale structures, complicate any comparison of the relaxation time between the three solutions.

Past the attenuation region, as the Reynolds number is further increased, we see another change in trend of turbulent intensity: the attenuation effect is diminished, and the turbulent intensity progressively returns to the value of the Newtonian fluid once again. This behavior is analogous to what has been observed in studies of drag reduction in channel flow [14, 16], where a critical Reynolds number exists past which the drag reduction effectiveness sharply declines and the flow returns to the Newtonian state. In such studies, the authors argue that while presence of some shear is essential for the formation of shear-induced-structures which play a key role in the drag reduction, excessive shear leads to the breakup of such structures and inhibit their formation, which undermines the drag reduction effectiveness.

While the current experimental arrangement differs

from the case of a channel flow, analogous conclusions can be drawn, and moreover, we can propose that this process happens in two main steps, linked to the distinct regimes of turbulence intensification and attenuation that have been reported. Initially, for very slow rotation rates, the mean shear favors the formation of very large SIS, which then interact with the forcing frequency leading to the previously discussed resonance phenomenon. As the rotation rate is increased, these structures break down, or their formation is simply not favored, but the individual string-like micellar structures remain intact. In the second regime, the turnover time of the smallest eddies surpasses the relaxation time of these small micellar structures, and the attenuation of turbulence beings. However, as the rotation rate is further increased, and we approach the threshold of $Wi = \mathcal{O}(10^2)$ in the bulk, the shear rate becomes large enough that these smaller micellar structures break down. The resulting smaller micelles should then have much faster relaxation times, and the turbulence attenuation mostly ceases.

Here it is also important to note that we believe that the geometry of the blades themselves might play an important role in the rotation rate at which the breakdown happens in both regimes. The rectangular cross-section of the blades, while being useful in the sense that it allows us to isolate the inertial component of the flow forcing, also generates regions of very high shear rate in the vicinity of the leading and trailing edges. This in turn means that the Weissenberg number in this region is considerably higher than that of the bulk and the breakdown of the micellar structures might be concentrated in this region, especially in the case of the second regime at very high rotation rates.

Another possible hypothesis for the behavior observed

in the loss of turbulence attenuation is that before the breakdown process begins, the string-like micellar structures are saturated (at maximum extension) and are therefore not able to absorb more energy, even though the energy input rate is being increased with the increase of the rotation rate. This hypothesis would then be consistent with the observation that the rotation rate that which the turbulent energy ratio becomes positive again is proportional to the concentration of the solutions, with this process happening first for the 33 ppm solution and being most delayed for the 100 ppm solution. Nevertheless, the unclear rheological characteristics of the solutions make it difficult to precisely pin down the mechanisms at play here.

A final point worth point out is that we believe that the breakdown of the large micellar structures being aligned with a Weissenberg number close to unity is a coincidence which is brought forth by the characteristic of the current experiment, and these phenomena don't necessarily need to be aligned. It is still unclear whether the resonant frequency of these very large scale structures is determined mostly by the geometrical constraints of the experimental device, or simply by the solution itself (most likely), and also how the ratio between the two observed characteristic times changes depending on concentration and solution chemistry.

V. CONCLUSIONS

Mention about changing shape of the blades, other surfactant types etc

Argue that different experimental devices or techniques might allow us to untangle the phenomena which are happening here.

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- [1] B. A. Toms, Some observations on the flow of linear polymer solutions through straight tubes at large reynolds numbers, *Proceedings of the 1st International Congress on Rheology* **2**, 135 (1948).
 - [2] J. L. Lumley, Drag reduction by additives, *Annual Review of Fluid Mechanics* **1**, 367 (1969).
 - [3] A. Groisman and V. Steinberg, Elastic turbulence in a polymer solution flow, *Nature* **405**, 53 (2000).
 - [4] Y.-B. Zhang, E. Bodenschatz, H. Xu, and H.-D. Xi, Experimental observation of the elastic range scaling in turbulent flow with polymer additives, *Science Advances* **7**, eabd3525 (2021).
 - [5] K. Fukushima, H. Kishi, H. Suzuki, and R. Hidema, Modification of turbulence caused by cationic surfactant wormlike micellar structures in two-dimensional turbulent flow, *Journal of Fluid Mechanics* **933**, A9 (2022).
 - [6] Y. Kawaguchi, J. Wei, B. Yu, and Z. Feng, Rheological characterization of drag-reducing cationic surfactant solution: Shear and elongational viscosities of dilute solutions, in *Volume 1: Fora, Parts A, B, C, and D*, *Fluids Engineering Division Summer Meeting* (2003) pp. 721–728.
 - [7] H.-x. Zhang, D.-z. Wang, W.-g. Gu, and H.-p. Chen, Effects of Temperature and Concentration on Rheological Characteristics of Surfactant Additive Solutions, *Journal of Hydrodynamics, Ser. B* **20**, 603 (2008).
 - [8] F.-C. Li, Y. Kawaguchi, B. Yu, J.-J. Wei, and K. Hishida, Experimental study of drag-reduction mechanism for a dilute surfactant solution flow, *International Journal of Heat and Mass Transfer* **51**, 835 (2008).
 - [9] S. Goto, Y. Saito, and G. Kawahara, Hierarchy of antiparallel vortex tubes in spatially periodic turbulence at high reynolds numbers, *Physical Review Fluids* **2**, 064603 (2017).
 - [10] T. Jahn, D. Schanz, and A. Schröder, Advanced iterative particle reconstruction for lagrangian particle tracking, *Experiments in Fluids* **62**, 179 (2021).
 - [11] L. Shindler, M. Moroni, and A. Cenedese, Spatial-temporal improvements of a two-frame particle-tracking algorithm, *Measurement Science and Technology* **21**, 115401 (2010).
 - [12] N. Wiener, *Extrapolation, Interpolation, and Smoothing*

- of Stationary Time Series* (MIT Press, Cambridge, MA, 1949).
- [13] P. Wang, S. Pei, M. Wang, Y. Yan, X. Sun, and J. Zhang, Study on the transformation from linear to branched wormlike micelles: An insight from molecular dynamics simulation, *Journal of Colloid and Interface Science* **494**, 47 (2017).
 - [14] M. Motozawa, T. Watanabe, W. Gu, and Y. Kawaguchi, Experimental Investigation on Zonal Structure in Drag-Reducing Channel Flow with Surfactant Additives, *Advances in Mechanical Engineering* **3**, 120438 (2011).
 - [15] A. K. M. F. Hussain and W. C. Reynolds, The mechanics of an organized wave in turbulent shear flow, *Journal of Fluid Mechanics* **41**, 241 (1970).
 - [16] J. Wei, J. Wang, C. Zhang, and Y. Kawaguchi, Combined effects of temperature and Reynolds number on drag-reducing characteristics of a cationic surfactant solution, *The Canadian Journal of Chemical Engineering* **90**, 1304 (2012).